

Mid-ocean ridges

The rock beneath Earth's crust is as hot as molten volcanic lava, but it is kept solid by intense pressure. When plates of ocean crust are pulled apart, rifts open up and reduce the pressure, allowing the hot rock to melt and shoot up through the rift. This creates chains of volcanoes that form long ridges of submarine mountains. These mid-ocean ridges are among the largest geological features on the planet.



RIFT ZONE

Where plates of oceanic crust are being dragged apart by convection currents in the hot mantle rock below, the ocean floor sinks to form a rift valley. The base of the valley is full of cracks that allow molten rock to erupt and form new ocean floor. Meanwhile, the heat raises blocks of ocean crust on each side of the rift valley to form a double ridge.

PILLOW LAVA

The rock that erupts through cracks in the ocean floor is molten basalt, like the lava that erupts from volcanoes in Hawaii. When it hits the cold water, it turns solid on the outside, but the molten rock keeps bursting out through the hard shell to form cushion-shaped mounds called pillow lava.

▼ RIDGES AND VALLEY

This cross section through a mid-ocean ridge shows how a rift in the ocean floor creates a valley with underwater mountains on each side of it.

